

AI-Enabled Elephant Detection and Train Alert System

Project Proposal - Group 10

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Introduction

The Conflict

- Human-elephant conflict is a critical safety and conservation issue in Sri Lanka.
- Wild elephants frequently cross railway tracks, leading to fatal accidents.
- In early 2025, a train strike near Polonnaruwa's Gal Oya junction killed seven elephants.
- While accounting for ~5% of elephant fatalities, these incidents are high-profile and largely preventable.



Problem Statement

The Problem

- In Sri Lanka, many elephants cross railway lines near their habitats, and trains often hit them, causing deaths and damage.
- Even with signboards and speed limits, drivers cannot see elephants in time.
- Unlike existing AI systems that rely only on camera detection and **internet-based alerts**
- So, we need a system that can detect elephants **in real-time** and warn the train driver early, even without **internet**.

TRAINS AND ELEPHANTS: MAY NEVER THE TWAIN SHALL MEET

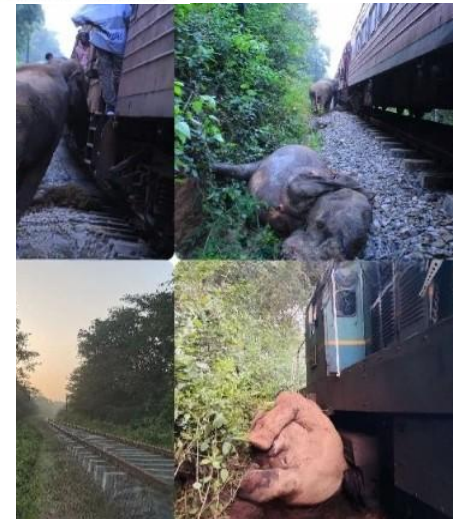


Six elephants dead after being hit by train in Sri Lanka

20 February 2025

Koh Ewe
BBC News

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Aims and Objectives

AIM: To develop and deploy a real-time, integrated early warning system to prevent train-elephant collisions and enhance railway safety along critical habitat zones.

To design: a multi-sensor detection module (camera, vibration, and acoustic) for identifying elephant presence near railway tracks.

To develop and train: an AI-based image recognition model (YOLO/CNN) to achieve over 90% detection accuracy with reduced false positives.

To implement: a LoRa-based wireless communication system to transmit real-time alerts to the train operator over 2-3 km distance without internet dependency.

To design and implement: Implement sensor fusion to reduce false positives and ensure reliability.

To evaluate: Prototype and test all detection, communication, and interface units.

Proposed Solution

An integrated system of sensors, AI, and wireless communication without relying on the internet

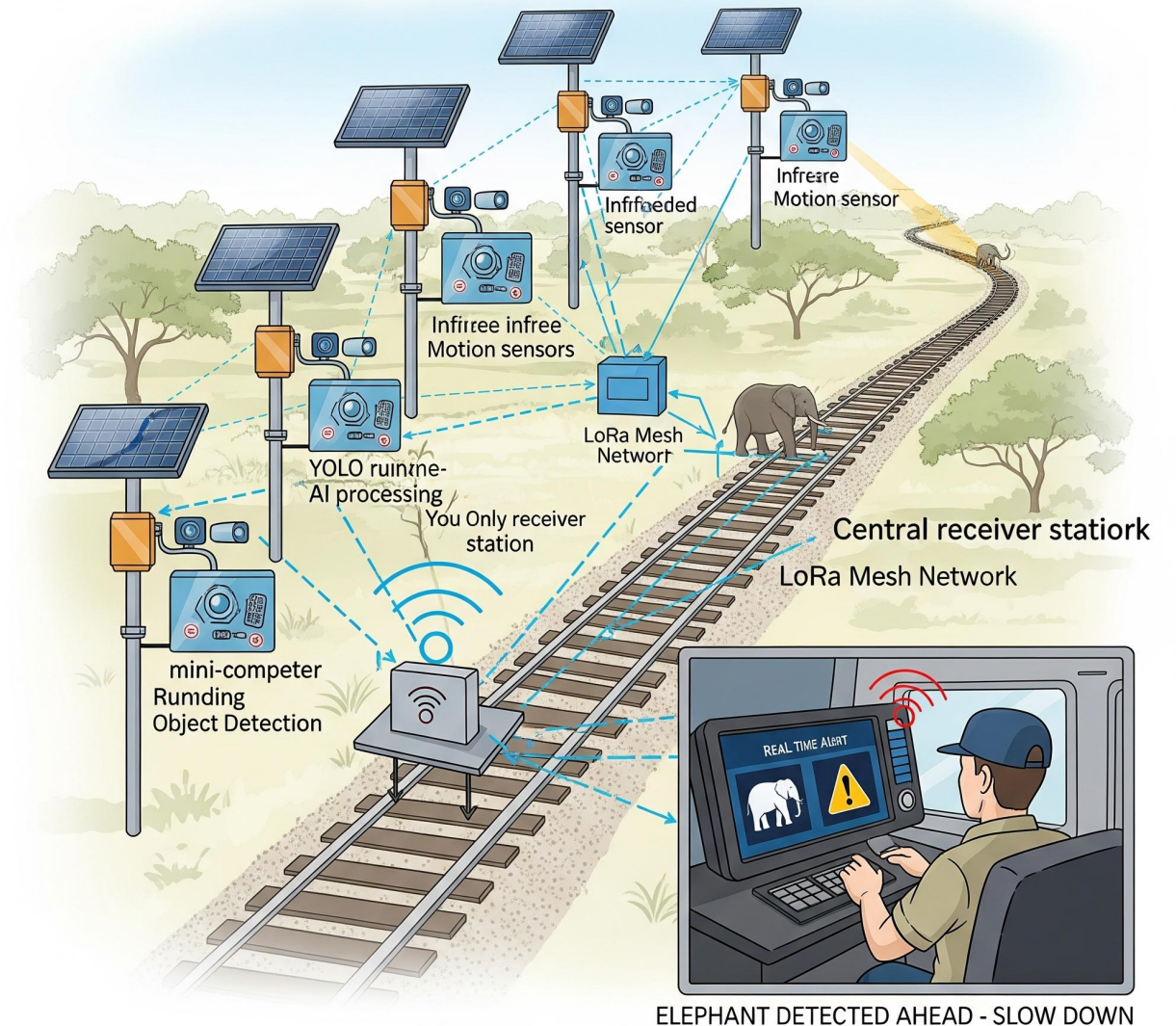
Detection: Use vibration, laser sensors identify elephants near railway tracks.

Verification: AI analyzes camera images to confirm the presence of elephants and reduce false alarms.

Communication: Send real-time alerts to trains using a LoRa-based wireless network **without internet** dependency.

Alerting: Equip trains with visual/audio warnings for immediate driver response if the train has entered the danger area.

Integrated System



Literature Review: Key Themes

Paper / Year	Technology Used	Limitations Identified	Solution / Improvement
A Convolutional Neural Network Based Early Warning System to Prevent Elephant-Train Collisions <i>(2021)</i>	• CNN-based visual detection (camera images) • Early warning app	• Depends on clear visual imagery • Night performance limited (unless IR/illumination) • Single-sensor vision → false positives	• Pi NoIR camera + IR illumination for night • Vibration (SW-420) + PIR/ultrasonic as pre-triggers • Lightweight CNN on Pi for local confirmation
Deep Vision Based Surveillance System to Prevent Train-Elephant Collisions <i>(Gupta et al., 2020)</i>	• Transfer-learning CNNs (ResNet/MobileNet/Inception) • Custom lightweight CNN	• High computational cost • Needs large labeled datasets • May not handle occlusion/foilage	• Lightweight model (MobileNet/YOLO-lite) • Sensor fusion (vibration + PIR/ultrasonic) • Pi runs inference only when target likely

Literature Review: Key Themes

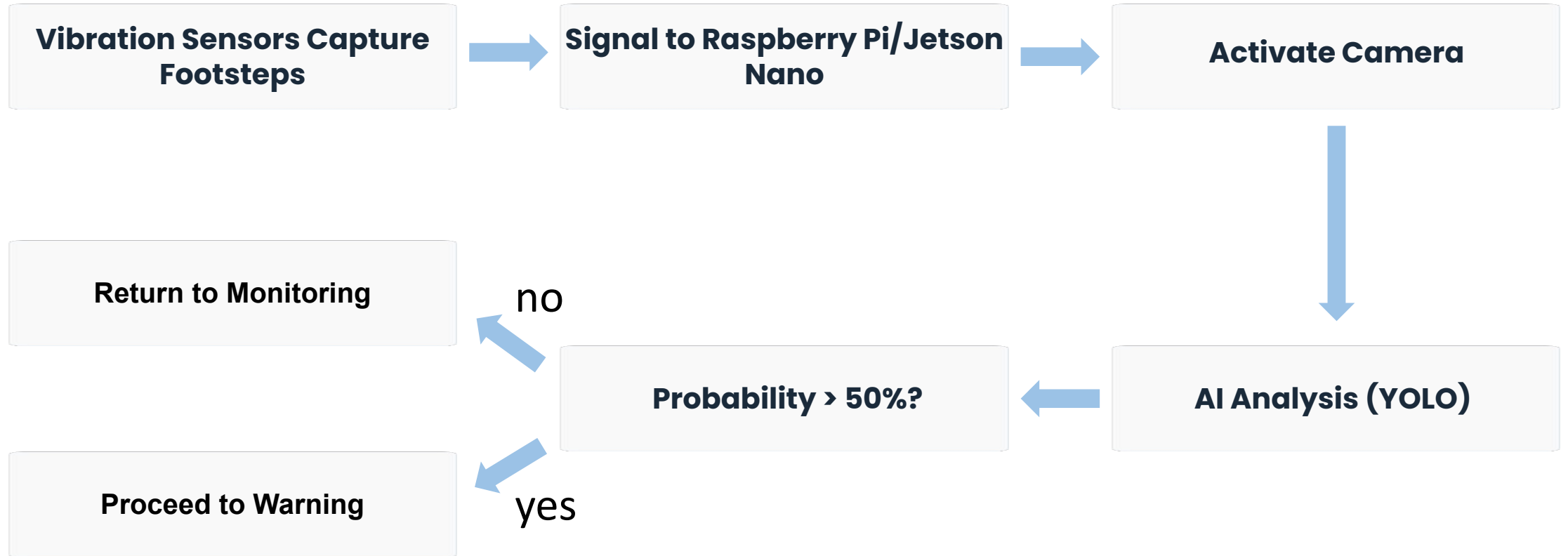
Paper / Year	Technology Used	Limitations Identified	Solution / Improvement
LoRa-Based Elephant Detection System Near Railroads <i>(2022-2024)</i>	• LoRaWAN / SX127x modules • Long-range alert transmission	• Many lack reliable confirmation → false alerts • Ignore antenna choice, placement, power	• On-node confirmation (camera+AI) • Proper Tx/Rx antenna selection • Clear receiver design • Solar/battery sizing
A Real-Time Thermal Image Processing Using Deep CNN for Monitoring Intrusion of Elephant <i>(2023-2024)</i>	• Thermal imaging • Deep CNN for night detection	• Thermal cameras expensive • Low-res thermopiles need interpolation • Many need high-res thermal for longer-range	• MLX90640 (32x24) as low-cost close-range thermal • Pi NoIR + IR LEDs for medium range • Fuse thermal + camera + vibration data

Literature Review: Key Themes

Paper / Year	Technology Used	Limitations Identified	Solution / Improvement
Photo-electric Infrared Beam Barrier Detection for Elephant Intrusion Monitoring <i>(~2021-2022)</i>	• Photo-electric IR beam barriers (1.75–2.20 m height)	• Detected 100% adults but only ~29% sub-adults • Limited sites • Sunlight/dust reduces reliability • Cannot identify species	• SEN1680 IR break-beam sensors at two heights (1m & 2m) • Combine with Pi NoIR + AI confirmation • Improves accuracy, identifies intruders

Methodology

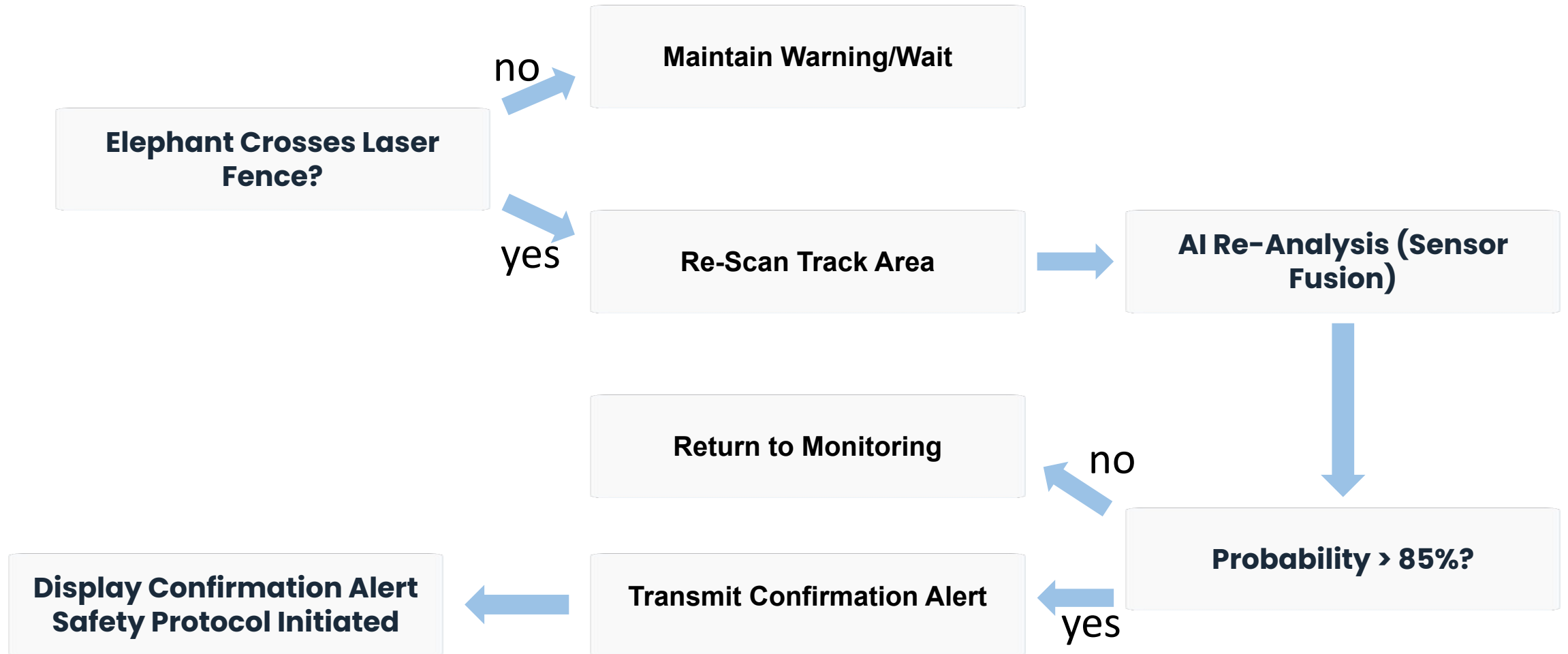
Phase 1: Early Detection (Perimeter Monitoring)



Phase 2: Network & Integration (Warning Transmission)



Phase 3: Confirmation Alert (High-Reliability Alert)



Project Budget Overview

- Estimated costs for building and testing the prototype system.

Component	Purpose	Approx. Cost (LKR)
Raspberry Pi 4 (4GB)	Main AI Processor	~30,000
Pi NoIR V2 Camera Module	Day / Night Vision Capture	~10,000
5W 850nm IR Light Module	Night-time Illumination	~3,500
Beam Sensor (SEN1680) * 2	Detects animal crossing a line	~5,400
LoRa Module (SX1278) * 2	Long-Range Alert Transmission	~4,800
Vibration Sensor (SW-420)	Detects Ground Tremors	~200
IP65 Waterproof Enclosure	Protects Electronics from Weather	~3,500
Power Supplies (Pi + IR)	Powers all components	~2,600
Wires, Mounts, Misc.	Assembly & Mounting	~1,000
	Total Estimated Cost	~70,000

Next Steps

Immediate actions to begin the project:



Finalize team roles and detailed work plan.



Secure components, and other resources.



Engage with Sri Lanka Railways and Wildlife Dept. for testing permissions.



Begin AI model training and sensor prototyping.



Maintain weekly milestones and documentation.

Conclusion

Our system is a technological early warning solution designed to stop train-elephant collisions, enhancing both railway safety and animal conservation by integrating AI and IoT.

This project demonstrates how technology can promote harmony between humans and wildlife.

References (Selected)

- A Convolutional Neural Network Based Early Warning System to Prevent Elephant-Train Collisions (2021) - [researchgate.net](https://www.researchgate.net)
- Deep Vision Based Surveillance System to Prevent Train-Elephant Collisions (Gupta et al., 2020) - assets-eu.researchsquare
- LoRa-Based Elephant Detection System Near Railroads (2022–2024) - seer-ufu-br.online
- A Real-Time Thermal Image Processing Using Deep CNN for Monitoring Intrusion of Elephant (2023–2024) - [researchgate.net](https://www.researchgate.net)
- Photo-electric Infrared Beam Barrier Detection for Elephant Intrusion Monitoring (~2021–2022) - onlinelibrary.wiley.com

Questions?

Thank you.